

- Big Crunch 75
 Big Dipper 102, 110
 Big Rip 75
 binary star systems 40, 42, **43**
 supernovas 34
 X-ray 114
 biosignature 83
 birth, of stars **30–1**
 BL Lacertae 129
 black dwarfs 28, 29, 33
 Black Eye Galaxy (M64) 138
 black holes 11, 18, 28, 29, 35, **38–9**, 125, 137
 in galaxies 50, 114, 199
 in Milky Way 54
 supermassive 38–9, 59, 60, 61, 116, 174, 184, 212
 blazars 60, 129
 Blaze Star (T Coronae Borealis) 117
 blue dwarfs 28
 blue giants 21, 25
 blue hypergiants 24
 Blue Planetary Nebula (NGC 3918) 177
 blue stars 20, 24, 25, 44, 133
 blueshift 70
 BM Scorpii 140
 Bode, Johann Elert 89
 Bok globules 30
 The Book of the Fixed Stars (al-Sufi) 89
 Boomerang Nebula 176, 177
 Boötes 112, **116–17**
 Brahe, Tycho 88, 89
 bright emission nebulae 163
 brightness
 dips in 42, 43, 46
 and size 25
 stars 20, 21, 22
 Brocchi's Cluster 148
 brown dwarfs 18, 25
 Bruno, Giordano 17
 "bubble Universes" 73
 Bug Nebula (NGC 6302) 141
 Bullet Cluster 75
 Butterfly Cluster (M6) 140, 141
 Butterfly Nebula see Bug Nebula
- C**
 Cacciatore, Niccolò 149
 Caelum **191**
 calendars 87
 Camelopardalis **109**
 Cancer **168–9**
 Canes Venatici **112–13**, 116
 Canis Major 94, 162, **194–5**
 Canis Minor 162, **169**
 Canopus (Alpha Carinae) 177, 194, 202, 203
 Capella (Alpha Aurigae) 132, 133
 Caph (Beta Cassiopeiae) 106, 107
 Capricornus 40, **186**
 carbon 29, 33
 and life 82
 Carina 196, 200, **202–5**
 Carina Nebula (NGC 3372) 18–19, 59, 202, 203
 dust clouds 204–5
 Cartwheel Galaxy (ESO 350–40) 190, 191
 Cassini space probe 83
 Cassiopeia **106–7**
 Cassiopeia, Queen 102, 106, 126
 Cassiopeia A 107
 Castor (Alpha Geminorum) 166, 167
 cataclysmic variables 42
 catalogues, star 88, 89
 Cat's Eye Nebula (NGC 6543) 104, 105
 Cebalrai (Beta Ophiuchi) 144
 Celeris 151
 celestial coordinates 91
 celestial equator 91, 93, 144, 154, 174
 charts centered on 96, 98–101
 celestial meridian 91
 celestial objects **18–19**
 celestial sphere **90–1**
 mapping the sky **94–5**
 and zodiac 92
 Centaurus **176–7**, 178
 Centaurus A (NGC 5128) 176, 177
 Center for Nuclear Research (CERN) 14
 Cepheid variables 42, 102, 146, 148, 167, 216
 Cepheus **102–3**
 Cepheus, King of Ethiopia 102, 106, 126
 Cepheus OB2 association 59
 Cetus 41, **158–9**
 Cetus (sea monster) 126, 130, 158
 Chamaeleon 188, **214**
 The Chamaeleon I Cloud 214
 Chandra X-ray Observatory 37, 80
 images from 80, 115
 Charles I, King of England 112
 charts, star 89
 Chelae Scorpionis 139
 Chertan (Theta Leonis) 135
 Chi Carinae 203
 Chi Eridani 160
 Chinese, ancient 89, 109, 183
 Chiron (mythological centaur) 177
 chromosphere 27, 227
 CI 0024+17 74
 Cigar Galaxy (M82) 110, 111
 Circinus **206**
 Circinus Galaxy 206
 the Cirlet 154
 circumpolar stars 96, 97
 civilizations, number of 83
 clocks 87
 clouds
 Neptune 231
 Venus 228
 clusters of galaxies see galaxy cluster
 CMBR see Cosmic Microwave Background
 Radiation
 Coalsack Nebula 178
 the Coathanger 148
 COBE satellite 17
 cold gas giants 46
 colliding galaxies **62–5**, 149, 174, 189
 Collinder 399 148
 collisions
 galaxies **64–5**, 66, 67
 galaxy clusters 75
 star 45
 color
 and size 24, 25
 spectral classification 20–1
 and temperature 20
- Columba **195**
 Coma Berenices 51, 68, 136, **138**
 Coma Cluster 66, 138
 coma (head) (comets) 18
 comets 18, 30, 223, 224, 225, 230
 and transfer of life 82
 Cone Nebula 170
 constellations
 Andromeda **126–7**, 150
 Antlia **199**
 Apus 188, **214**
 Aquarius **152–3**
 Aquila **146–7**
 Ara **182**
 Aries 92, **157**
 Auriga **132–3**
 Boötes 112, **116–17**
 boundaries 94, 95
 Caelum **191**
 Camelopardalis **109**
 Cancer **168–9**
 Canes Venatici **112–13**, 116
 Canis Major 94, 162, **194–5**
 Canis Minor 162, **169**
 Capricornus 40, **186**
 Carina 196, 200, **202–5**
 Cassiopeia **106–7**
 Centaurus **176–7**, 178
 Cepheus **102–3**
 Cetus 41, **158–9**
 Chamaeleon 188, **214**
 changing shape of 106
 charting the heavens 88–9
 Circinus **206**
 Columba **195**
 Coma Berenices 68, 136, **138**
 Corona Australis **183**
 Corona Borealis **117**, 183
 Corvus **174**, 175
 Crater **175**
 Crux **178–9**, 200
 Cygnus 40, 41, **124–5**
 Delphinus **149**
 Dorado 188, **210–11**
 Draco **104–5**, 118
 Equuleus **151**
 Eridanus **160–1**
 Fornax **192–3**
 Gemini 92, **166–7**
 Grus **188–9**
 Hercules 88, **118–19**
 Horologium **218**
 Hydra 95, **172–3**, 175, 217
 Hydrus 188, **217**
 Indus 188, **208**
 Lacerta **129**
 Leo 51, **134–5**
 Leo Minor 88, **132**
 Lepus **193**
 Libra 136, **139**
 locators 97
 Lupus **180**
 Lynx **108**
 Lyra **120–1**
 mapping the sky **94–5**
 Mensa **219**
 Microscopium **189**
- Constellations (continued)
 Monoceros **170–1**
 Musca 188, **206**
 Norma **181**
 number of 87, 94
 Octans **219**
 Ophiuchus 92, 142, **144–5**
 Pavo 188, **216**
 Pegasus 51, 68, 89, **150–1**
 Perseus 106, **130–1**
 Phoenix 188, **209**
 Pictor **212**
 Pisces 51, **154–5**, 157
 Piscis Austrinus 152, **187**, 188, 189
 Puppis **196–7**, 200
 Pyxis **198**
 Reticulum **213**
 Sagitta **149**
 Sagittarius **184–5**, 189
 Scorpius 92, **140–1**, 162, 181
 Sculptor **190–1**
 Scutum **147**
 Serpens 30, **142–3**
 Sextans **174**
 sky charts 96–101
 Taurus 88, 89, **156–7**
 Telescopium **207**
 Triangulum 13, **128**
 Triangulum Australe 188, **207**
 Tucana 188, **215**
 Ursa Major **110–11**, 112
 Ursa Minor **102**, 110
 Vela 196, **200–1**
 Virgo 43, 50, 92, **136–7**
 Volans 188, **213**
 Vulpecula 125, **148**
 zodiacal 92
 convection 26
 convective zone 27
 coordinates, celestial 91
 Copernicus, Nicolaus 17
 Cor Caroli (Charles's Heart) (Alpha Canum Venaticorum) 112, 113
 core
 dying stars 32
 end of fusion in 34
 following supernova explosions 29
 Mercury 228
 stars 20, 26, 27, 28
 supergiant 34
 Uranus 231
 Whirlpool Galaxy 114, 115
 corona
 stars 27
 Sun 226
 Corona Australis **183**
 Corona Borealis **117**, 183
 coronal mass ejections 227
 Coronet Cluster 183
 CoRoT spacecraft 48
 Corvus **174**, 175
 cosmic expansion 70–1
 and dark energy 75
 cosmic golden egg 16
 Cosmic Microwave Background Radiation (CMBR) 16, 72
 variations in 17